EC 441 — Midterm Exam Reference Sheet

Spring 2026 Boston University

Packet Delays

$d_{trans} = L/R$ (L bits, R b/s link rate)
$d_{prop} = d/v$ (d distance, v prop. speed)
$v \approx 2 \times 10^8$ m/s (copper/fiber), 3×10^8 m/s (free space)
End-to-end over N store-and-forward links:
$d_{total} = N \cdot d_{trans} + N \cdot d_{prop}$

Information Theory

Self-information: $I(p) = -\log_2 p$ bits
Shannon entropy: $H(X) = -\sum_i p_i \log_2 p_i$ bits/symbol
Max entropy (uniform): $H_{max} = \log_2 M$ for M symbols

SI and Binary Prefixes

Name	SI	Binary
Kilo/Kibi	10^3	$2^{10} = 1024$
Mega/Mebi	10^6	$2^{20} \approx 1.05 \times 10^6$
Giga/Gibi	10^9	$2^{30} \approx 1.07 \times 10^9$
Tera/Tebi	10^{12}	$2^{40} \approx 1.10 \times 10^{12}$

Decibels and Power

$A[\text{dB}] = 10\log_{10}(P_{out}/P_{in})$
$P[\text{dBm}] = 10\log_{10}(P[\text{mW}]/1\,\text{mW})$
$P[\text{dBW}] = 10\log_{10}(P[\text{W}]/1\,\text{W})$
Useful values: $+3$ dB $\approx 2\times$; -3 dB $\approx 0.5\times$; $+10$ dB $= 10\times$

Shannon-Hartley Capacity

$$C = B\log_2\left(1 + \frac{S}{N}\right) \quad \text{b/s}$$

B = bandwidth (Hz), S/N = linear SNR
$\text{SNR[linear]} = 10^{\text{SNR[dB]}/10}$

Guided Media (typical)

Medium	Bandwidth	Range
Twisted pair	1–1000 MHz	∼100 m
Coaxial	1–1000 MHz	∼500 m
Optical fiber	> 10 THz	100+ km

Digital Signaling

Bit rate = symbol rate $\times \log_2 M$ (M symbol levels)
Bipolar NRZ BER (matched filter, AWGN): $P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$
$Q(x) = \frac{1}{\sqrt{2\pi}}\int_x^\infty e^{-u^2/2}\,du$
Doubling M costs ≈ 6 dB extra E_b/N_0 for same BER.

Wireless: Free-Space Path Loss

Friis equation: $P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$
$L_{FSPL}[\text{dB}] = 32.45 + 20\log_{10}(f_{\text{MHz}}) + 20\log_{10}(d_{\text{km}})$
Link budget (dB/dBm/dBi):
$P_r = P_t + G_t + G_r - L_{FSPL} - L_{misc}$

QAM Spectral Efficiency

Modulation	Bits/symbol	Min SNR (approx)
BPSK	1	6 dB
QPSK	2	9 dB
16-QAM	4	16 dB
64-QAM	6	22 dB
256-QAM	8	28 dB

Cellular Frequency Reuse

Reuse distance: $D = R\sqrt{3N}$
C/I approximation: $C/I \approx (3N)^{n/2}/6$
N = reuse factor, R = cell radius, n = path-loss exponent

Error Control Coding

(n, k) block code: k data bits, n-bit codeword
Code rate: $R_c = k/n$
d_{min} theorem: $e_c + e_d \le d_{min} - 1$, $e_c \le e_d$
Detect only: $d_{min} \ge e_d + 1$
Correct only: $d_{min} \ge 2e_c + 1$
Single-bit correction bound: $n + 1 \le 2^{n-k}$
Frame error probability (independent bit errors):
$P_{frame} \approx 1 - (1-p)^n \approx np \quad (p \ll 1)$

CRC (Modulo-2 Arithmetic)

Encoding: (1) append r zeros; (2) divide by $G(x)$ (mod 2) $\to R(x)$; (3) send $T(x) = x^r M(x) + R(x)$
Check: divide received frame by $G(x)$; remainder $= 0 \Rightarrow$ OK
CRC-32 ($r = 32$): used by Ethernet, WiFi, ZIP.
Detects all bursts ≤ 32 bits; all single, double, odd-bit errors.

ALOHA Throughput

Protocol	Throughput S
Pure ALOHA	$S = Ge^{-2G}$, max $\approx 18.4\%$ at $G = 0.5$
Slotted ALOHA	$S = Ge^{-G}$, max $\approx 36.8\%$ at $G = 1$

G = offered load (frames/frame-time or frames/slot)

CSMA/CD

Min frame size: $L_{min} = 2\tau R$ (τ = max prop. delay, R = bit rate)
Standard: 64 bytes at 10 Mb/s over 2500 m.
Binary exponential backoff: after k-th collision, $r \in \{0, 1, \ldots, 2^{\min(k,10)} - 1\}$ slot times.
Give up after 16 collisions.

Ethernet Frame

Field	Content	Size
Preamble	`10101010` × 7	7 B
SFD	`10101011`	1 B
Dst MAC	destination address	6 B
Src MAC	source address	6 B
Type/Len	EtherType or length	2 B
Payload	data	46–1500 B
FCS (CRC-32)	frame check sequence	4 B

Min/max frame (excl. preamble+SFD): 64 B / 1518 B

Selected EtherTypes

`0x0800` = IPv4 `0x0806` = ARP `0x86DD` = IPv6

MAC Addresses

48-bit; first 3 bytes = OUI (manufacturer).
Bit 0 of first byte: 0 = unicast, 1 = multicast.
Broadcast: `FF:FF:FF:FF:FF:FF`

ARP Protocol

Request: broadcast "Who has IP x? I am `<my-ip>/<my-mac>`."
Reply: unicast "IP x is at MAC `<mac>`."
Off-subnet: ARP for the *default gateway*, not the destination.

Switch Forwarding

Destination MAC	Action
Known unicast	Forward to recorded port
Unknown unicast	Flood all except ingress
Broadcast / multicast	Always flood

Learn source MAC on every received frame. Entries expire (∼300 s).

Ethernet Generations (selected)

Standard	Speed	Medium	Line code
10BASE-T	10 Mb/s	Cat 3 UTP	Manchester
100BASE-TX	100 Mb/s	Cat 5 UTP	4B5B+MLT-3
1000BASE-T	1 Gb/s	Cat 5e UTP	4D-PAM5
10GBASE-T	10 Gb/s	Cat 6A UTP	—

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